

Accessing GRAPHICAL DESKTOP Of RASPBERRY Pi Using SSH And VNC

KALPESH BHOSLE

Raspberry Pi is a low-cost, high-performance, credit-card-sized computer for learning, solving problems and having fun. Easy to get started with, this tiny yet powerful computer comes with additional resources including monitor screen, keyboard and mouse. Imagine the inconvenience of moving around the accessories between workplaces for accessing this tiny computer. Does it still feel tiny?

The configuration given in this article lets you set up a local network

between Raspberry Pi board and a laptop using an Ethernet cable. It is a simple and convenient configuration to access Raspberry Pi remotely.

Things you need to get started

There are three things needed to configure the setup: Raspberry Pi with an SD card running Raspbian operating system (OS), laptop running Linux OS—I prefer Ubuntu 16.04—and Ethernet cable to set up a connection between Raspberry Pi and laptop (Fig. 1).

The laptop must have an Internet connection to install a few software packages. The same connection can be shared with Raspberry Pi using an Ethernet cable to install required packages.

Installing required packages on Ubuntu

Configure Ethernet connection on Ubuntu to share its network with hosts connected to it—in this case, Raspberry

Pi. Then, open connection settings for Ethernet connection from desktop.

Change IPV4 settings to Share to Other Computers and click on Save. (Fig. 2). An IP address is required to get access to Raspberry Pi over Secure Socket Shell (SSH). (SSH is a network protocol that provides the administrator with a secure way to access a remote computer. SSH service is not enabled by default in Raspberry Pi. To enable SSH, access command line using `sudo raspi-config` command). Since there is no monitor or keyboard connected to Raspberry Pi, it is not possible to use `$ ifconfig` command.

To find out the IP address assigned to Raspberry Pi using a laptop, use `nmap` tool. Ubuntu needs Internet connectivity over Wi-Fi. To do that, connect it to a known Wi-Fi connection.

Open the terminal and install `nmap` using the following command:

```
$ sudo apt-get install nmap
```

Once the installation is complete, reboot the system. Make sure it is connected to the Internet via Wi-Fi.

Accessing Raspberry Pi using SSH

Connect Raspberry Pi to the computer with Ubuntu using an Ethernet cable. Then, start Raspberry Pi. By default, the network configuration saved in Raspberry Pi over Ethernet connection is `dhcp`. This allows Raspberry Pi to get the IP address from Ubuntu system.

It takes some time for Raspberry Pi to boot up and get an IP address. Once it gets an IP address, a pop up on Ubuntu system will display Connection Established.

Note down the broadcast address of the Ethernet connection on Ubuntu. Open the terminal and run `$ ifconfig` command as shown in Fig. 3. Make a note of the IP address of Ubuntu system (10.42.0.1 is the IP address in this example) and look for the term

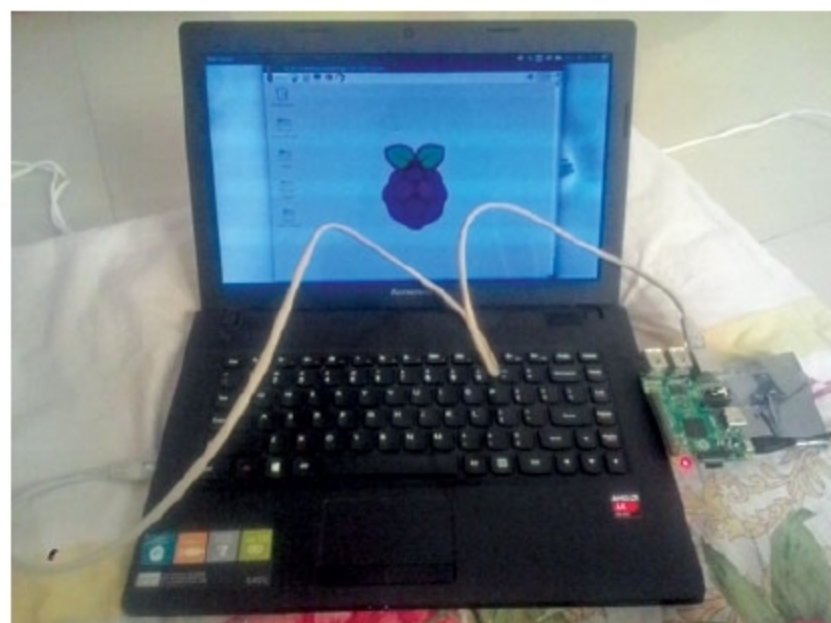


Fig. 1: Raspberry Pi setup on laptop using Ethernet cable

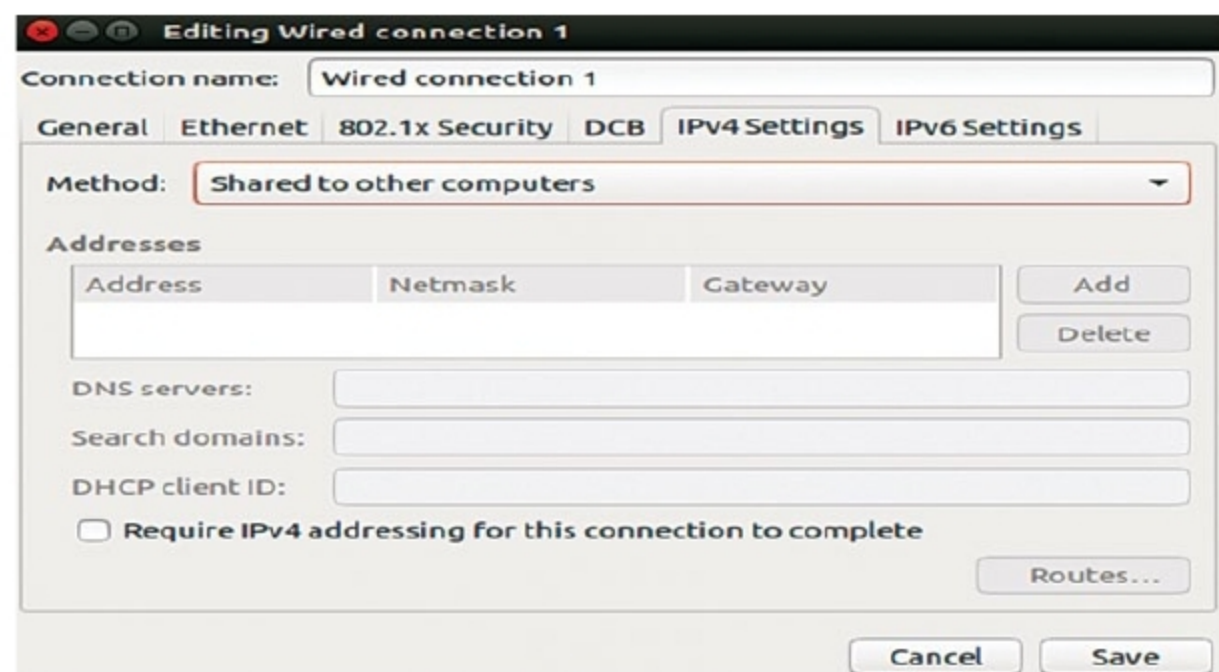


Fig. 2: Ethernet settings in Ubuntu